

A Comprehensive Report on My Capstone Project Analysis Findings

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OUTLINE



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- Methodology
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- Discussion
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EXECUTIVE SUMMARY



Summary of key findings

- Python is going to be the most in-demand language next year.
- JavaScript is the most worked with language at this year.
- PostgreSQL is the most worked with database this year and will be the most in-demand database next year.
- AWS is the most worked with cloud platform this year and will be the most in-demand cloud platform next year.
- The USA holds majority of the job market for developers.
- It is worth noting that I completed the entire project as required.



INTRODUCTION



- The purpose of this report is to present my capstone project findings.
- My target audience is my grading instructor at IBM
- The analysis contains key findings with relation to how people utilise certain software, how their experience affects their compensation and much more.
- The data used was extracted from the Stack Overflow Developer Survey.
- The Value of the analysis is to know the current trends and keep up with the upcoming trends for developers.



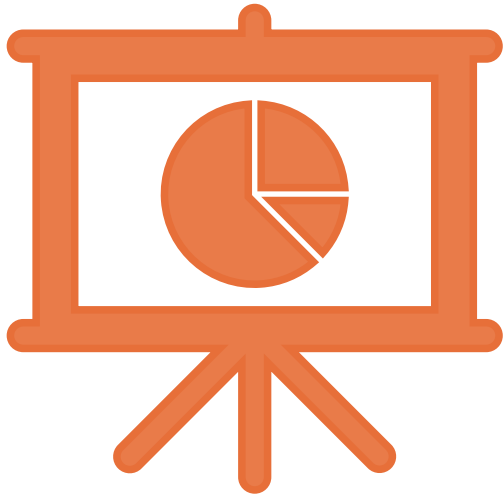
METHODOLOGY



Data sources, Data collection & Data wrangling steps

- The data used was extracted from the Stack Overflow Developer Survey.
- Data collection was done using API access and web scrapping techniques.
- Data Wrangling was carried out on the data such as:
 - Identifying and removing duplicate rows
 - Handling missing values
 - Examining employment status for value counts
 - Normalising the data for comparative analysis.

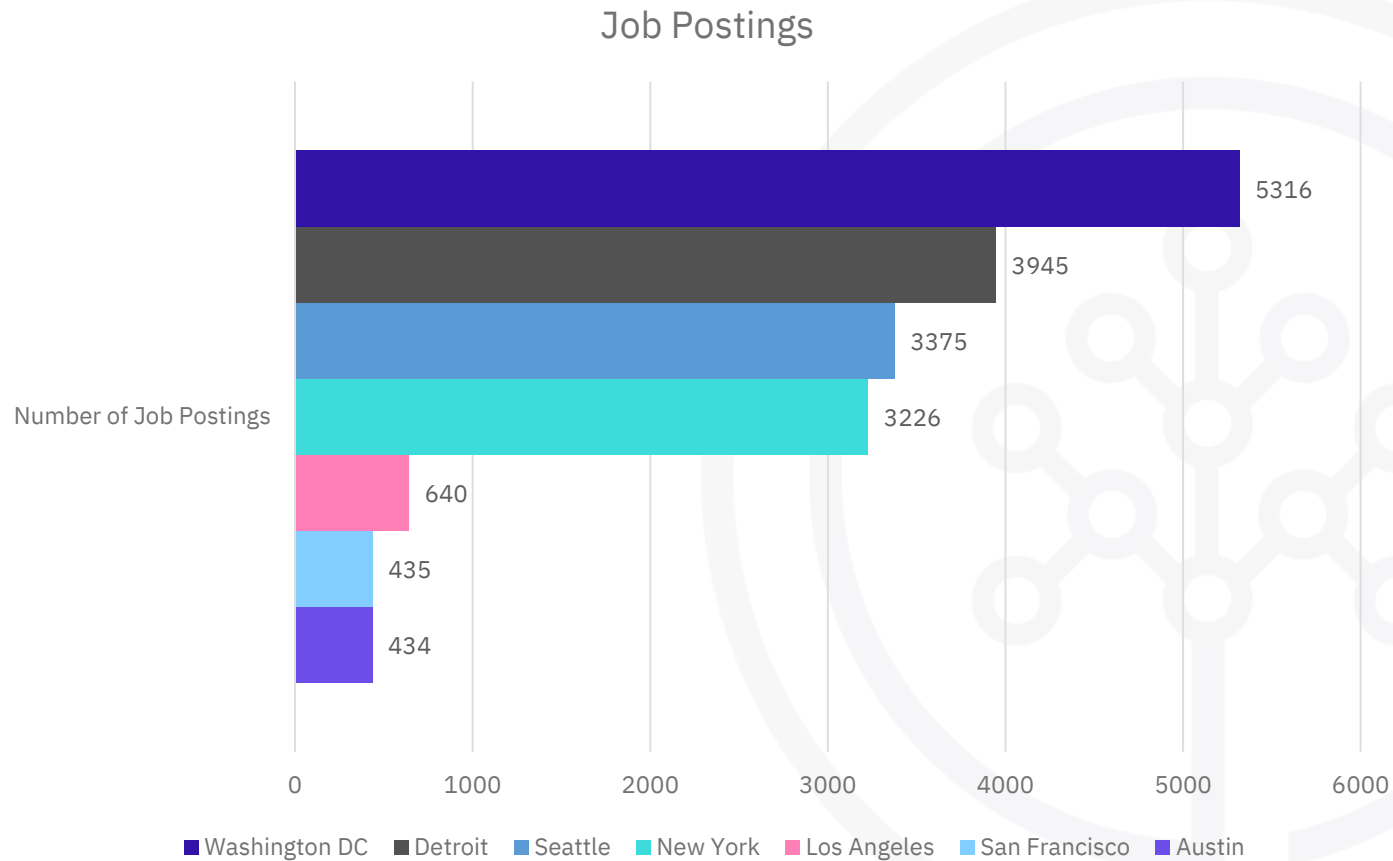
Results



<Please present your results in the following slides.>

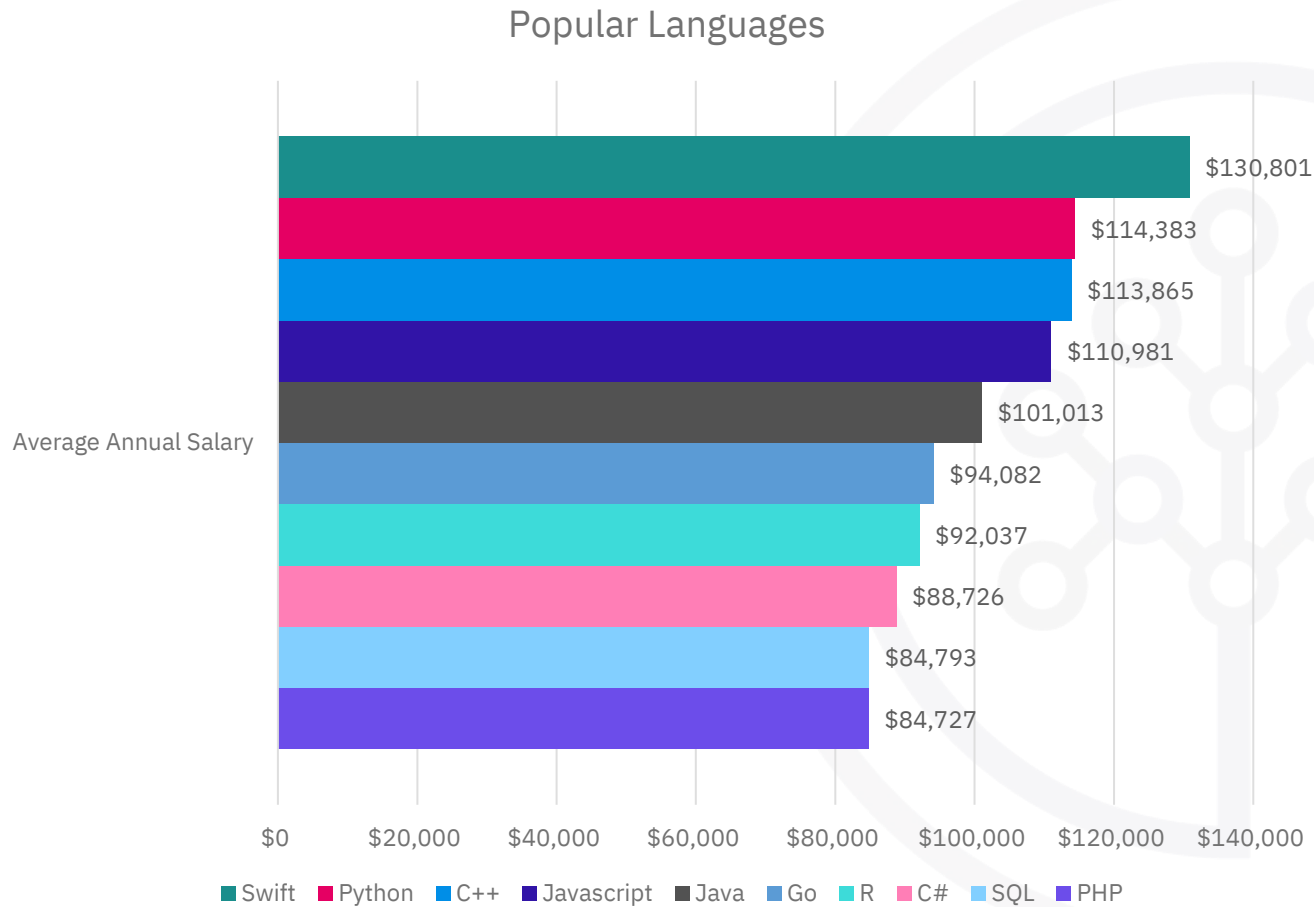


JOB POSTINGS



- From the chart to the left, we can clearly see that Washington DC has the highest amount of Job Postings (5,316 persons) with Austin being the lowest (434).
- It's clear that the job market is in DC

POPULAR LANGUAGES



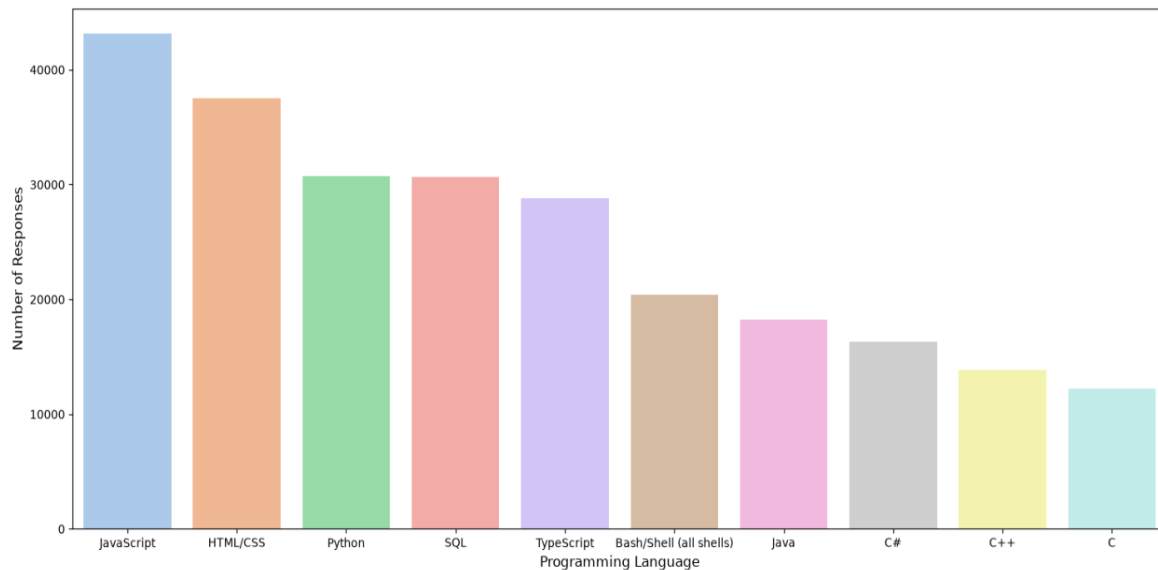
- From the bar chart to my left, we can see that the highest paying Programming Language on average is Swift (\$130,801) with the lowest being PHP (\$84,727)

PROGRAMMING LANGUAGE TRENDS

Summarize key trends shown in the charts.

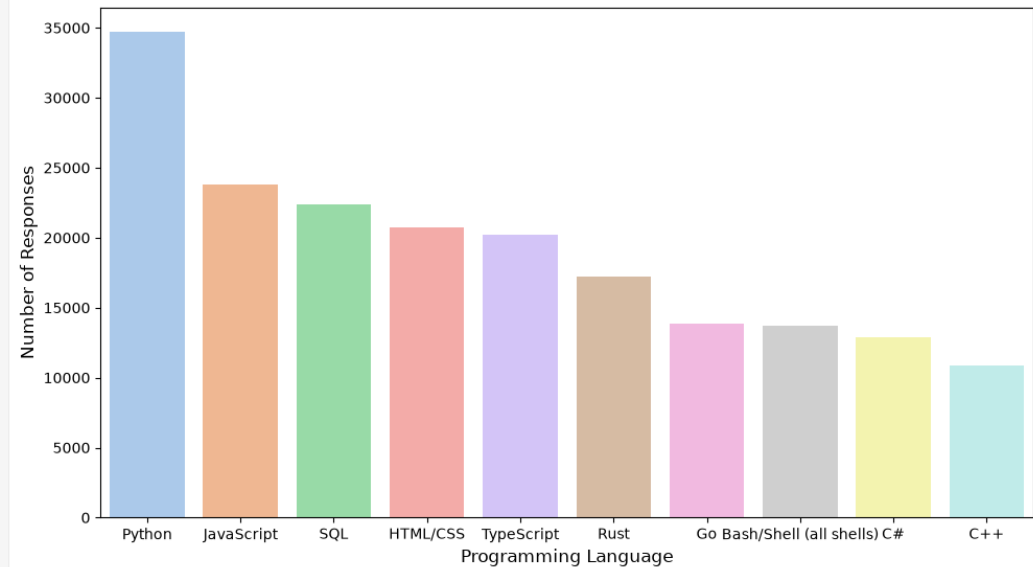
Current Year

Top 10 Most used Programming Languages



Next Year

Top 10 Most Desired Programming Languages



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript is the most used language in the current year
- Python is the language most people want to work with next year while the desire to learn C++ is relatively low.
- Languages such as Python, SQL and Rust have moved up the pile with JavaScript still being wanted but not as before.

Implications

- Currently every website runs with JavaScript or at least one of its frameworks and it makes sense for it to be in high demand.
- The rush for python indicates that people see the language as being in demand in the coming year.
- The rush for Python and SQL can be attributed to the rise in AI and the demand for data professionals to feed AI models. These languages form the core of machine learning

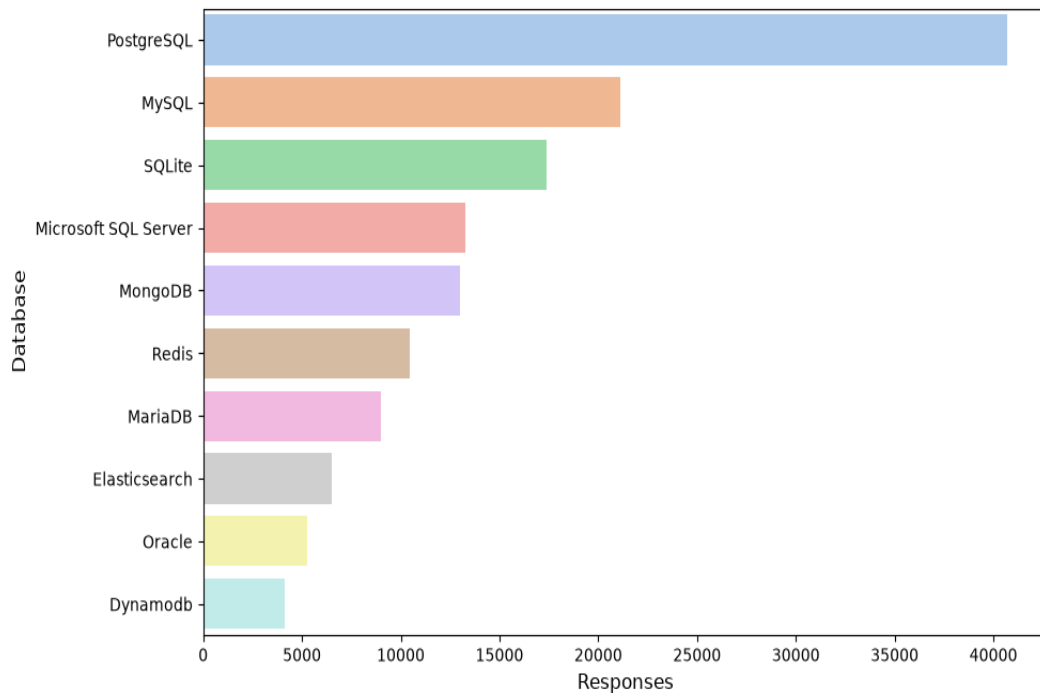


DATABASE TRENDS

Summarize key trends shown in the charts.

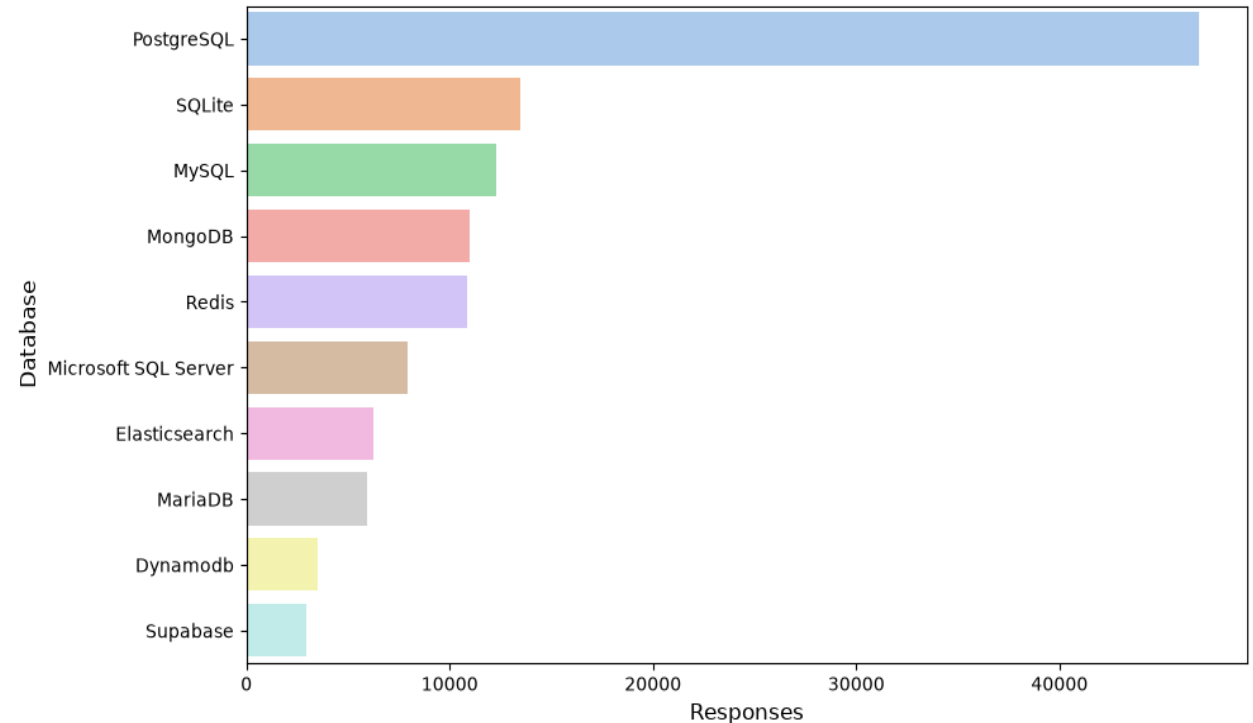
Current Year

Top 10 Most Worked with Databases



Next Year

Top 10 Most Desired Databases



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is the most worked with database of the current year.
- PostgreSQL is still the most desired database of next year with SQLite moving up the pile but in the desired area.
- PostgreSQL is clearly a power house in its field with Oracle losing its edge over time.

Implications

- Developers see PostgreSQL as the best choice for the database needs in the current year.
- PostgreSQL is still the most desired database. This further asserts its dominance in the database world.
- It is clear that PostgreSQL is the way to go for future database managers. It is very scalable and trusted by big companies.

DASHBOARD



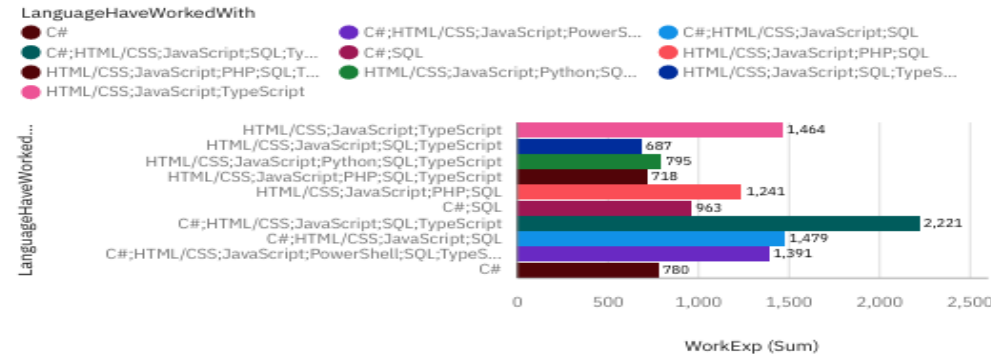
<Please present your dashboard in the following slides.>



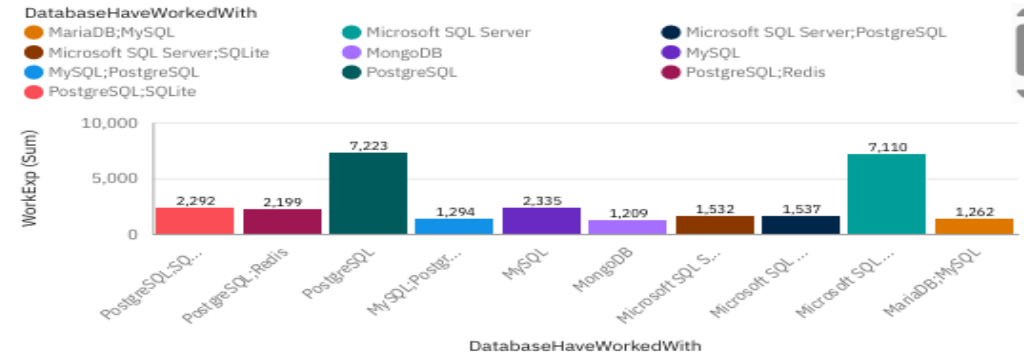
DASHBOARD TAB 1: Current Technology Usage

Current Technology Usage

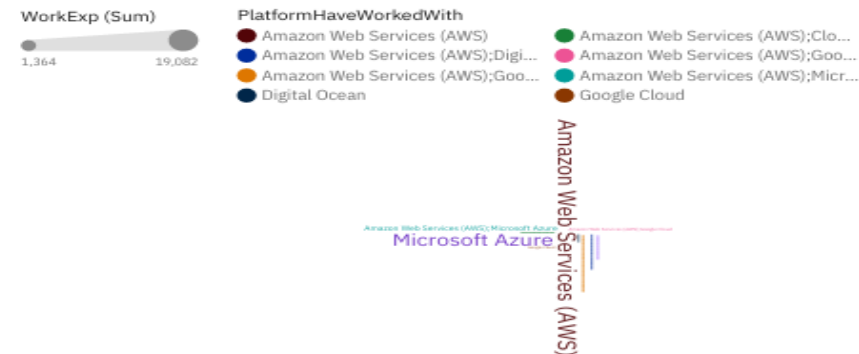
Familiar Language against Work Experience



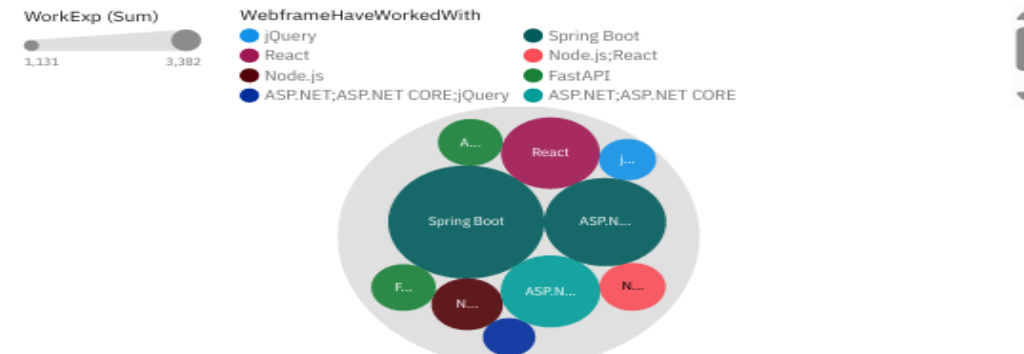
Familiar Database against Work Experience



Familiar Platforms against Work Experience



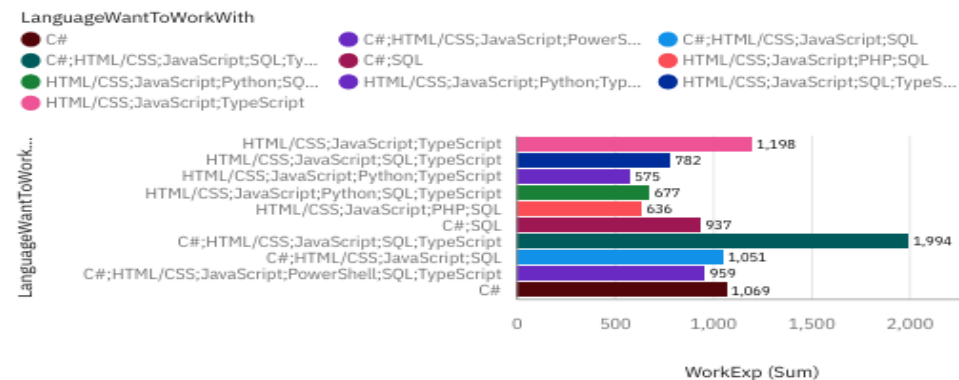
Familiar Web Frameworks against Work Experience



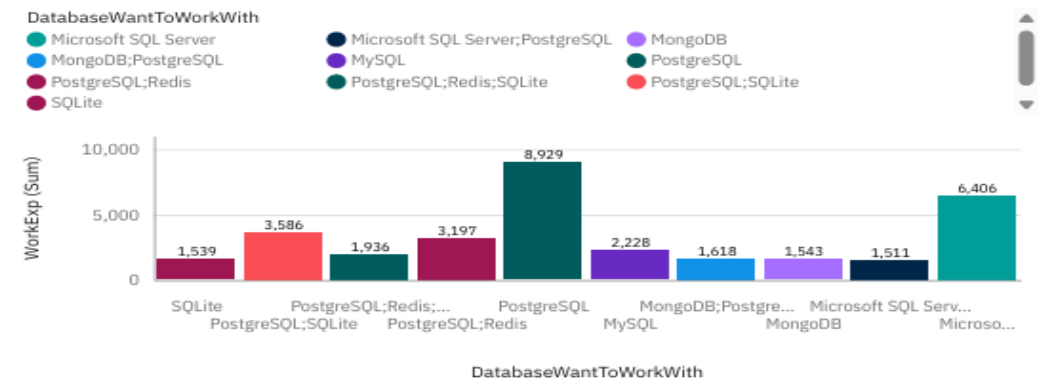
DASHBOARD TAB 2: Future Technology Trends

Future Technology Trend

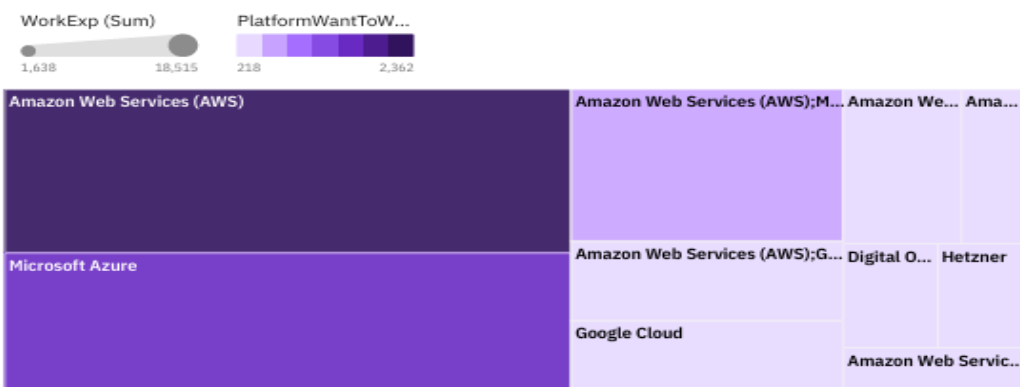
Desired Language against Work Experience



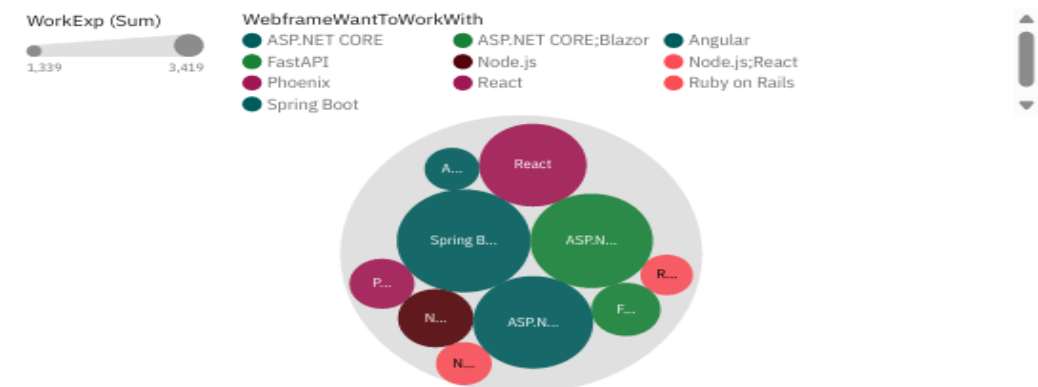
Desired Database against Work Experience



Desired Platform against Work Experience



Desired Web Framework against Work Experience



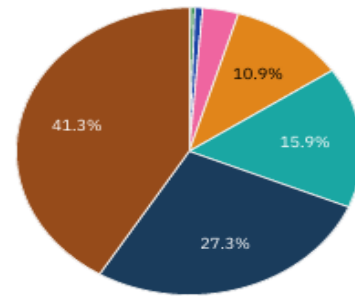
DASHBOARD TAB 3: Demographics

Demographics

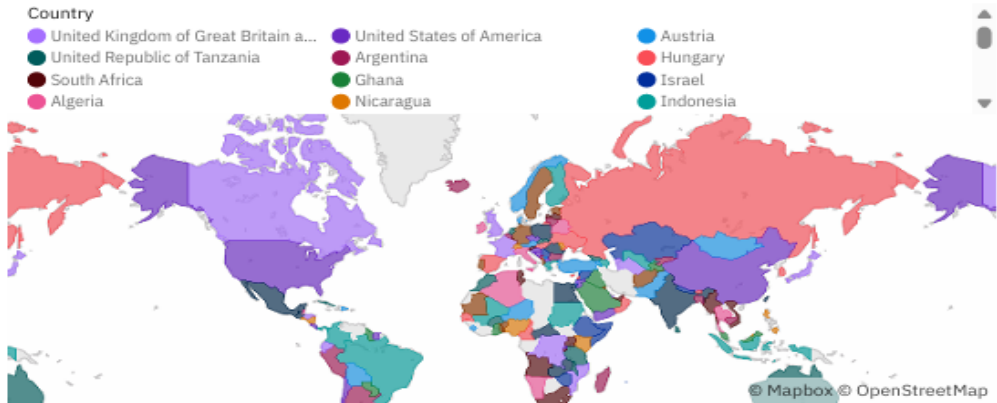
Respondent's Distribution by Age

Age

- Prefer not to say
- 18-24 years old
- 25-34 years old
- 35-44 years old
- 45-54 years old
- 55-64 years old
- 65 years or older



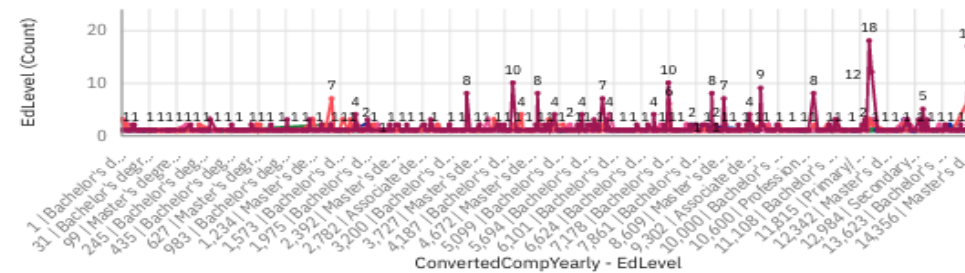
Respondent's Count by Country



Respondent's Distribution by Education Level

EdLevel

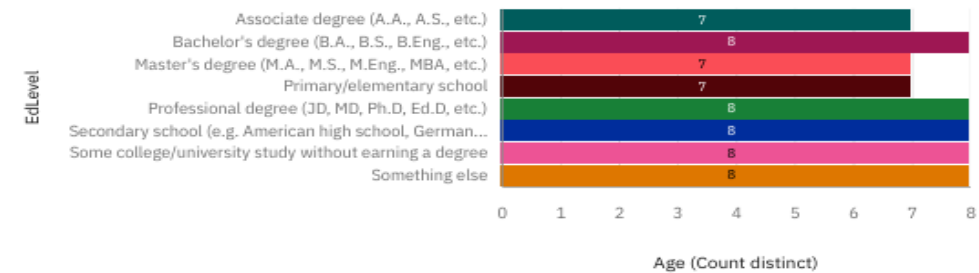
- Associate degree (A.A., A.S., etc.)
- Primary/elementary school
- Some college/university study wit...
- Bachelor's degree (B.A., B.S., B.E...
- Professional degree (JD, MD, Ph.D...
- Something else
- Master's degree (M.A., M.S., M.En...
- Secondary school (e.g. American ...



Respondent's Age by Education Level

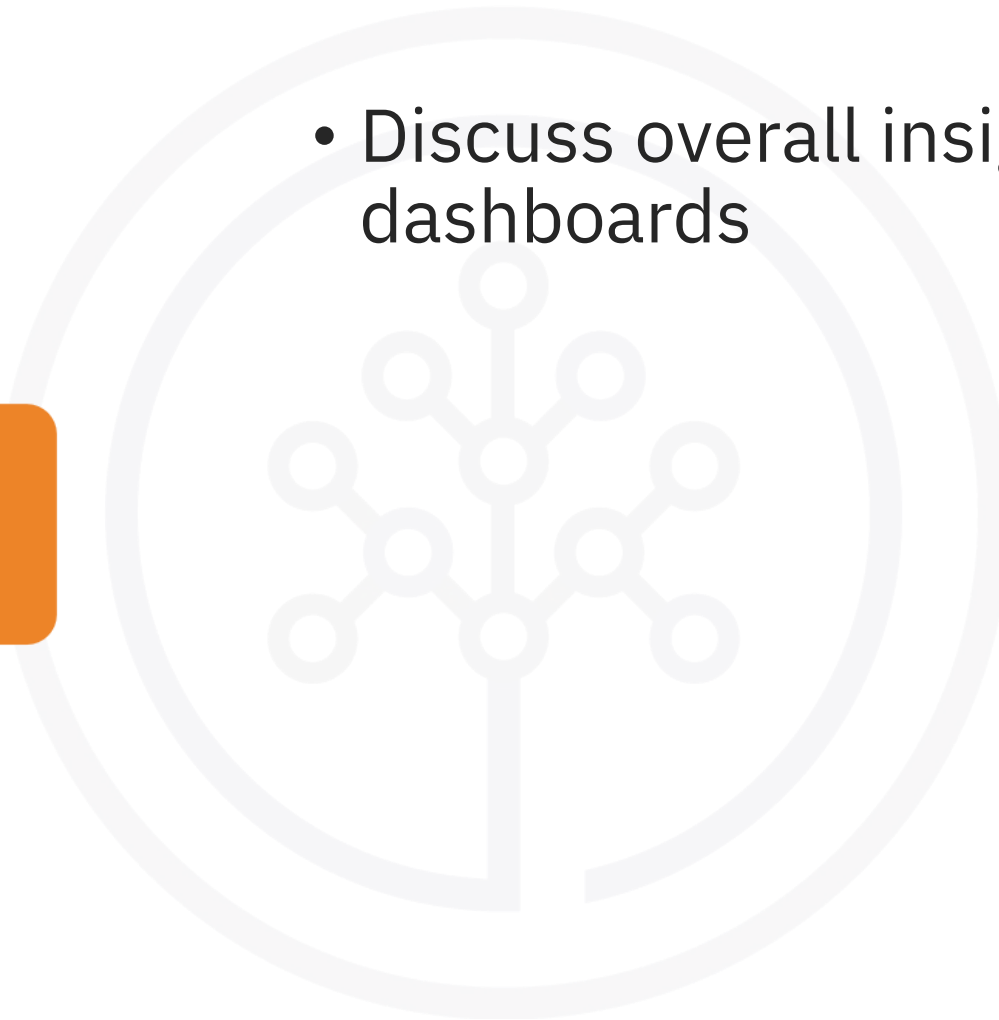
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DISCUSSION

- Discuss overall insights derived from dashboards



OVERALL FINDINGS & IMPLICATIONS

Findings

- From the first dashboard, it is clear that the current leading language is JavaScript, database is PostgreSQL, cloud platform is AWS and web framework is Spring Boot
- From the second dashboard the desire for JavaScript is still top in languages (though it has dropped), PostgreSQL has since a rise in its desire, AWS stands strong amongst cloud platforms and Spring Boot is still top for desired frameworks though it seems to have dropped.
- From the third dashboard most respondents who provided their ages are of the range 35-44 years of age, Majority of them are from the United States, with many having at least an elementary level of education

Implications

- These four hold the current developer market with a strong presence in each of their respective fields
- The hierarchy stayed very much the same with the difference being the number of respondents in each case.
- This shows that Americans of age 35-44 years form the majority of developers in the world. This means the job market is highly competitive in that area.



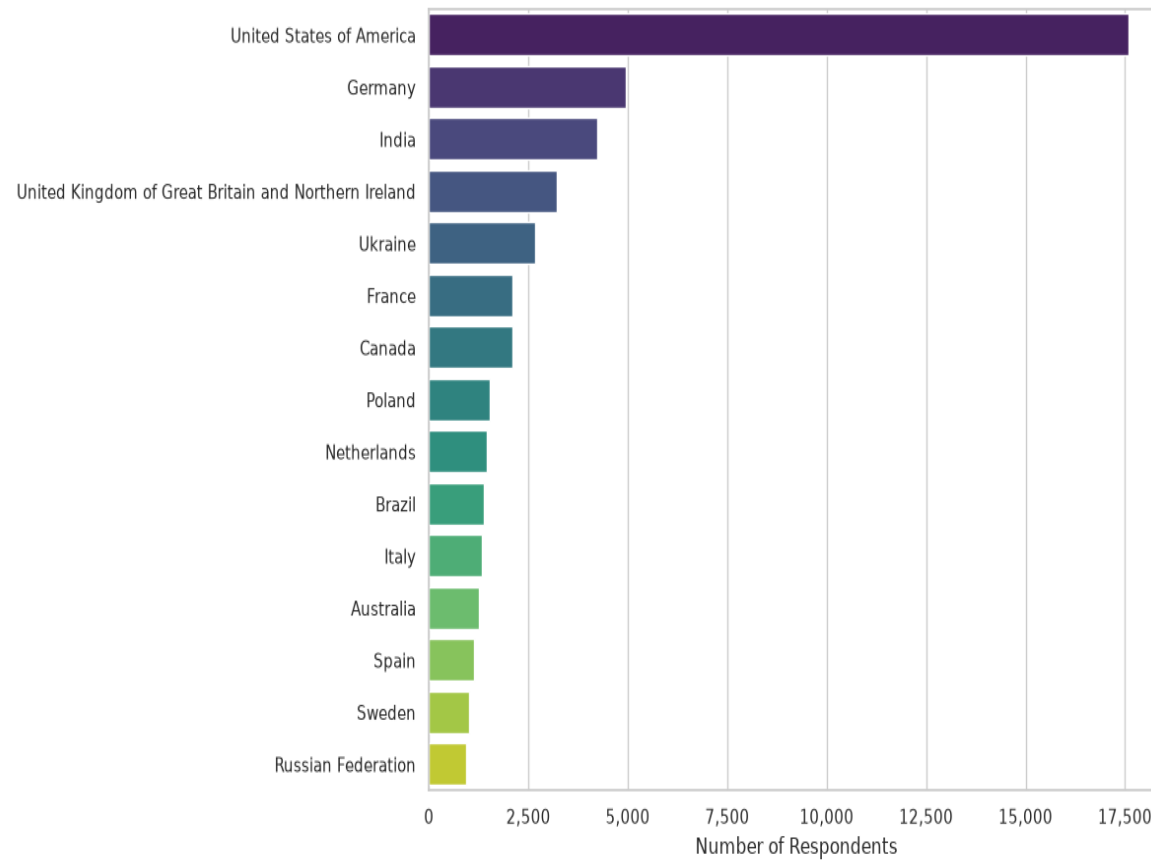
CONCLUSION



- From the analysis I can conclude that the most ideal language to learn as a developer next year will be Python because the world is greatly leaning towards that direction with the coming of AI. SQL is also an essential language.
- PostgreSQL has been and will keep being the way to go database for developers. Its scalability and ease of use keep driving developers and companies towards it.
- AWS is the leading cloud platform. In a world where everything can be found on the cloud, acquiring this skill will be highly beneficial for any developer.
- Finally, it is clear that the USA holds the developer job market and people with high educational qualifications have an advantage. It will be wise to attain relevant educational qualifications so as to stand out amongst employers

APPENDIX

Top 15 Most Represented Countries by Respondent Count



- This graph shows just how dominant the USA is when it comes to Technology in the world.
- It more than doubles its follower, Germany.